

WASP-43b

R-0388

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_250310 · created 2026-07-18T23:09:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460744.869 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.027
Transit depth (Rp/Rs)^2	2.97 +/- 0.93 [%]
Orbital Inclination (inc)	81.28 +/- 2.05
Airmass coefficient 1 (a1)	0.984 +/- 0.018
Airmass coefficient 2 (a2)	0.0084 +/- 0.0084
Scatter in the residuals of the lightcurve fit is	1.79 %
Best Comparison Star	#2 - [259, 80]
Optimal Method	PSF photometry
Transit Duration (day)	0.0475 +/- 0.0096

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.172 ± 0.027 vs 0.1594 (deviation 0.47σ)
Duration fit vs published	0.0475 d vs 0.0512 d (deviation 0.39σ)
Mid-time O–C	6.6 min
Depth SNR	6.4
Residual scatter	1.79 %

Light curve

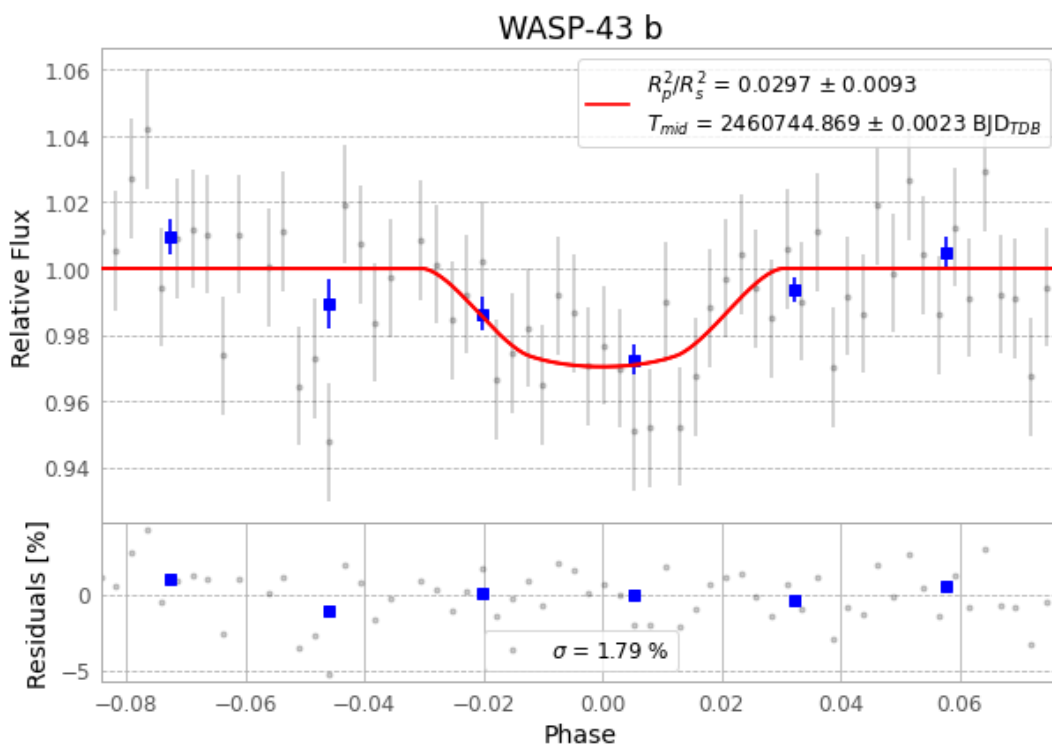


Figure 1 — FinalLightCurve_WASP-43 b_2025-03-10.png (SHA-256 e7015e58f572d32a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [379, 262], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aef54ca9a559dc7e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8350c6b

Data provenance

64 FITS frames (42 MB), WASP-43250310070316.FITS → WASP-43250310101215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250310.log (05086b22498f...), FinalLightCurve_WASP-43 b_2025-03-10.png (e7015e58f572...), FinalLightCurve_WASP-43 b_2025-03-10.csv (563818f669a5...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 23:09Z.